Intermediate.txt

```
AD 1 C 100

IS 3 1 S 0

IS 1 2 S 1

IS 2 1 S 2

DL 2 C 1

DL 1 C 5

DL 2 C 1

AD 2
```

Symtab.txt

```
0 A 103

1 B 104

2 C 105
```

Littab.txt

```
import java.io.*;
import java.util.*;

class Main {
    public static void main(String args[]) throws Exception {

        BufferedReader im = new BufferedReader(new
FileReader("Intermediate.txt"));
```

```
        BufferedReader st = new BufferedReader(new
FileReader("Symtab.txt"));
        BufferedReader lt = new BufferedReader(new
FileReader("Littab.txt"));
        PrintWriter out = new PrintWriter("Output.txt");

        int S[] = new int[10], L[] = new int[10];
        String line;
        int i = 0, j = 0;

        // Read Symbol Table
        while ((line = st.readLine()) != null) {
            StringTokenizer t = new StringTokenizer(line);
            t.nextToken();
            t.nextToken();
            S[i++] = Integer.parseInt(t.nextToken());
        }

        // Read Literal Table
        while ((line = lt.readLine()) != null) {
            StringTokenizer t = new StringTokenizer(line);
            t.nextToken();
            t.nextToken();
            L[j++] = Integer.parseInt(t.nextToken());
        }

        int address = 0;

        // Pass 2
        while ((line = im.readLine()) != null) {
            StringTokenizer t = new StringTokenizer(line);
            String word = t.nextToken();

            // Assembler Directive
            if (word.equals("AD")) {
                int code = Integer.parseInt(t.nextToken());

                if (code == 1) {
```

```
            t.nextToken(); // Skip C
            address = Integer.parseInt(t.nextToken());
        }
        continue;
    }

    // Imperative Statement
    if (word.equals("IS")) {
        int opcode = Integer.parseInt(t.nextToken());
        int reg = 0, operand = 0;

        if (t.hasMoreTokens()) {
            String x = t.nextToken();

            if (x.matches("[1-4]")) {
                reg = Integer.parseInt(x);

                if (t.hasMoreTokens())
                    x = t.nextToken();
            }

            if (x.startsWith("S"))
                operand = S[Integer.parseInt(x.substring(1))];

            else if (x.startsWith("L"))
                operand = L[Integer.parseInt(x.substring(1))];
        }

        String result = address + "   " +
            String.format("%02d", opcode) + "   " +
            reg + "   " + String.format("%03d", operand);

        System.out.println(result);
        out.println(result);
        address++;
    }

    // Declarative Statement
```

```
            else if (word.equals("DL")) {
                int code = Integer.parseInt(t.nextToken());

                if (code == 1) {
                    String result = address + "  00  0  000";
                    System.out.println(result);
                    out.println(result);
                    address++;
                }

                else if (code == 2) {
                    t.nextToken(); // Skip C
                    int size = Integer.parseInt(t.nextToken());
                    address += size;
                }
            }
        }

        im.close();
        st.close();
        lt.close();
        out.close();

        System.out.println("Pass 2 completed");
    }
}
```